Warming Trends: Analyzing the Impact of Temperature Anomalies on the Risk and Return of the US Stock Market

Master's Thesis

Amirmahdi Shokatabadi

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1st supervisor: Dr. Thomas Walther
2nd supervisor: Prof. dr. Frank Verbeeten

Utrecht School of Economics

Universiteit Utrecht

Abstract

In this study, I examine how temperature anomalies affect the stock returns and volatility of US public companies, in particular those in heat-sensitive industries. I use panel regressions with fixed effects that combine climate data with financial data for about 18,000 firms that are located and traded in the continental US. I also test whether a possible effect comes from the impact of temperature on firms' infrastructure, measured in the city of the headquarters, or from its effect on investor sentiment, measured in the city of the exchange. I find no evidence that temperature anomalies affect the returns of US stocks. However, I find some evidence that they may affect the standard deviation of stock returns.

Keywords: Temperature anomaly, US, Climate risk, Investor sentiment, Heat-sensitive industries

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1 Introduction

According to the IPCC's Sixth Assessment Report (IPCC, 2022), climate change at current levels of warming already has widespread and permanent effects on ecosystems and human communities. Warming of 1.5 degrees Celsius will bring more severe climate risks, including more extreme weather events and rising sea levels, with serious consequences for natural and human systems. The report stresses the need for immediate and sustained action to limit warming to this level, and for strong adaptation measures to deal with the impacts that can no longer be avoided. There is therefore an urgent need for a global response that reduces these risks and protects future generations. Lau and Nath (2012) analyze heat waves in North America and identify regions, such as the Great Plains and the West Coast, where temperatures change significantly. They expect the severity, duration, frequency and geographical extent of heat waves to increase in the 21st century compared with the 20th century.

Even though there is a lot of evidence that climate change is happening, it has remained a subject of debate in the US, especially between Republicans and Democrats. The debate is about whether climate change harms different parts of the US economy, and whether the laws that respond to it, such as limits on carbon emissions, hurt the economy. The conflict grew when Donald Trump withdrew the US from the Paris Agreement, arguing that it would cause losses for US companies. McCright and Dunlap (2011) show that for liberals and Democrats, higher education goes together with stronger agreement with the scientific consensus and more concern about climate change, while for conservatives and Republicans these factors have a weaker or even negative effect. According to Ongena and Dursun-de Neef (2024), people who experience warmer-than-usual temperatures become more concerned about climate change and move their deposits away from banks that finance fossil fuels. This effect is stronger in counties with more climate change deniers, which they measure by the share of Republican voters. Views on climate change in the US therefore differ strongly between groups, and research in this field can help investors and policymakers better understand how people form beliefs about global warming and how they react to it.

A broad literature studies how high temperatures affect the economy. Dell et al. (2012) show that temperature changes have a large effect on economic growth. High temperatures reduce growth in poor countries, where even a 1°C increase in temperature can lower growth substantially. Rich countries such as the US are much less affected. One reason may be that wealthier countries have stronger systems that help them cope with and adapt to changes in temperature. Jones and Olken (2010), however, show that rising temperatures may also hurt rich countries indirectly, through their imports from poor countries. Makridis and Schloetzer (2023) find that extreme temperatures (below 15 and above 85 degrees Fahrenheit) affect people's beliefs about current and future economic conditions in the US. This effect on economic sentiment is stronger among highly skilled people whose jobs have high cognitive demands.

Pankratz et al. (2023), who study more than 17,000 firms in 93 countries, find that firms earn lower revenues and operating income when they are exposed to more extremely hot days, and that capital markets do not fully anticipate the impact of heat on corporate profits. Addoum et al. (2020) study how temperature affects the sales and productivity of US companies, both at the firm and at the establishment level, and find no significant effect. However, in a later study, Addoum et al. (2023) show that extreme temperatures affect earnings in

more than 40 percent of industries, and that these effects are not limited to agriculture. These studies suggest that the ability of countries and industries to deal with the financial costs of climate change depends on their economic situation. However, researchers still disagree about the sign and the size of the effect.

Pankratz et al. (2023) and Addoum et al. (2023) also study how analyst forecasts and earnings announcement surprises are related to heat exposure. Pankratz et al. (2023) find that announcement returns become more negative as heat exposure increases. They conclude that analysts and investors do not fully take the effects of high temperatures into account, and that the effect is strongest after two quarters. Addoum et al. (2023), on the other hand, show that in the US, analysts' forecasts account for temperature shocks by the end of the quarter in many industries, but not in all of them, so in some industries temperature shocks can still lead to earnings surprises. They also find that analysts' forecasts and stock prices do not respond immediately to observable temperature shocks within the quarter.

There is also a line of behavioral finance research that studies how weather, rather than climate, affects investors' decisions. These studies look at factors such as sunlight, daily temperature and cloud cover. Cao and Wei (2005) find that temperature is negatively related to stock returns in many markets. They explain that lower temperatures can make investors more aggressive and more willing to take risks, which leads to higher returns, while at high temperatures apathy dominates. Their study shows how environmental factors such as temperature may influence investor behavior and market dynamics. Hirshleifer and Shumway (2003) examine the relationship between sunny weather and stock returns in 26 countries and find that sunshine is positively related to returns, which suggests that sunlight improves investor mood. This is in line with the wider evidence on how weather and mood affect decision-making. Saunders (1993) also supports this view: he finds that sunny weather in New York City is related to higher stock prices. Investor mood, which is affected by the weather, can therefore have an important effect on market movements. Kamstra et al. (2003) study Seasonal Affective Disorder (SAD) and show that changes in investor mood caused by changes in daylight lead to a seasonal pattern in stock returns. These studies challenge the idea that financial markets are driven only by rational decisions, and suggest that changes in mood caused by the weather can have a large impact on markets. Since sunlight and temperature are closely related, behavioral studies of weather are close to the literature on climate risk, and the two can be studied together. Most behavioral papers measure the effect of temperature on investor sentiment through the location of the stock exchange. I follow this approach to assess the effect of temperature on investor sentiment.

I also pay special attention to heat-sensitive industries, because studies on climate change stress their role. Many studies have examined the effect of extreme temperatures on labor productivity and sales in these industries. For example, Graff Zivin and Neidell (2014) find that in the US, workers in climate-exposed industries work fewer hours on days when the maximum temperature exceeds 85°F, while people who are not working move from outdoor to indoor leisure on such days. This points to a loss of labor supply on hot days. Schlenker and Roberts (2009) study the effect of temperature on agricultural output in the US and provide strong evidence that climate change will harm agriculture. They show that crop yields increase with temperature up to a certain threshold, but fall sharply above it. Fisher et al. (2012) also find that climate change could have a severe impact on US agriculture, which is highly affected by temperature and precipitation. More

recent studies, such as Addoum et al. (2020) and Addoum et al. (2023), analyze how extreme temperatures affect the sales, productivity and earnings of establishments and firms, including those in these industries. Addoum et al. (2023) find that many of the industries whose earnings react to extreme temperatures are the high-exposure industries defined by Graff Zivin and Neidell (2014), although most of the affected industries are in consumer sectors. In contrast, Addoum et al. (2020) find no evidence that temperature extremes have a significant effect on establishment sales in heat-sensitive industries in the US. Overall, these studies show that the effect of high temperatures on heat-sensitive industries, such as agriculture, is an important question.

I include precipitation only as a control variable, because the leading studies find that it plays a smaller role than temperature. For example, Jones and Olken (2010) find that precipitation has a weaker effect on exports than temperature, and that this effect is not robust. Dell et al. (2012) find that precipitation has a mild effect on national growth, but in rich countries most of this effect comes from a few very large outliers. According to Dell et al. (2014), precipitation does not affect economic activity as much as temperature does. Addoum et al. (2020) include precipitation as a control variable, which shows that they see it as secondary when they study the effect of temperature on firm performance. Pankratz et al. (2023) also focus mainly on temperature, so the direct effect of precipitation on financial performance has received less attention. The existing studies give a mixed picture: the effect of precipitation on firm performance seems to differ by region and industry, with significant effects in some industries, and its direction, positive or negative, is still unclear. The behavioral finance studies of Saunders (1993), Hirshleifer and Shumway (2003) and Kamstra et al. (2003) suggest that weather has a broader effect on investor behavior, but they focus on sunshine, cloud cover and daylight rather than on precipitation. Since the effect of precipitation on economic and financial variables is complex, I also include it as a control variable.

Whether and how temperature affects the stock market is still debated, and studies disagree on what drives the effect. Makridis and Schloetzer (2023) find that the effect of extreme temperatures on sentiment spreads to asset prices, so that the stock returns of local firms fall on very cold or very hot days. They suggest that more frequent temperature extremes can increase non-fundamental risk in the market, and argue that extreme local temperatures affect people's beliefs beyond their effect on firm-level or local economic variables. In contrast, Apergis (2023) finds that high temperatures lower stock prices around the world, and attributes this to the effect of high temperatures on firms' assets, investments and borrowing costs.

This study adds to the literature by examining how the size of monthly temperature anomalies, that is, the magnitude of the deviation from the long-run average, affects stock returns and volatility, in particular for heat-sensitive industries such as agriculture, mining, transportation, utilities and construction. More specifically, I test whether the stocks of US firms, as firms in an advanced economy, are affected by temperature variation, and whether such an effect works through the impact of climate on the firms themselves or through the impact of weather on investor sentiment.

The work of Winter and Kiehl (2022) supports the use of temperature anomalies. They argue that anomalies reflect ongoing shifts in climate patterns better than static averages. Anomalies come with more frequent and more intense extreme weather events, which have immediate and severe economic consequences, while average temperatures can hide these fluctuations and their effects. When temperatures move away from

historical norms, societies have to adapt again, and this is costly. The costs include changes to infrastructure, new agricultural practices and new health measures to cope with extreme weather.

My research differs from existing studies in two ways. First, studies such as Addoum et al. (2020) and Pankratz et al. (2023) focus on extreme temperature events and measure temperature shocks as the number of days above or below a threshold, while I use the size of the deviation from the long-run average. Second, these studies examine how temperature affects firm performance, such as sales and labor productivity, or how equity analysts react to high temperatures. To my knowledge, the size of temperature anomalies, as a measure of climate variability, has not been directly related to the return and risk of the US stock market.

Market dynamics are closely related to investor sentiment. Most studies look at this from a behavioral finance perspective and focus on weather variables, such as sunshine and cloud cover, that affect investors' mood, rather than on climate variables as environmental risks. I go one step further and combine the effect of temperature in exchange cities on investor mood with the effect of climate on firms' infrastructure. This allows me to test whether possible changes in the return and risk of US public companies come from climate risk or from investor sentiment.

I use climate variables at the headquarters to measure the effect of climate risk on the stock market through firms' operations and infrastructure. This choice is based on the Efficient Market Hypothesis (Fama, 1970), which states that stock prices reflect all available information. If temperature shocks at a firm's headquarters affect the firm, this information should show up in its stock return and risk. I therefore examine the effect of temperature shocks at the headquarters on stock returns and risk.

Among these climate and behavioral studies, fewer look at volatility than at returns. Narayanamoorthy et al. (2015) use GARCH models to study the effect of weather on stock volatility in India. They find that temperature affects stock returns in some cities and volatility in others, and they stress that volatility should be taken into account when studying the relationship between weather and financial markets. Kathiravan et al. (2017) also find that temperature in several large Indian cities affects the returns of the main Indian stock indices, and they study its effect on volatility as well. They use descriptive statistics, the ADF test and GARCH(1,1) models.

Tzouvanas et al. (2019) use a ΔCoVaR model to study the effect of temperature fluctuations on the systemic risk of European firms. They find that temperature has a significant effect on systemic risk: hot temperature shocks increase systemic risk, while cold shocks reduce it. Their results show that the effect of temperature on systemic risk is asymmetric and nonlinear. Although the results differ somewhat across industries, they suggest that temperature variation plays an important role for systemic risk in European firms.

These advanced models are complex and beyond the scope of this research. To provide a broad analysis of the effect of temperature anomalies on the US stock market, I therefore use the monthly standard deviation of daily returns as a simple measure of volatility, and I study it for temperature anomalies in both the headquarters and the exchange cities.

I believe that my results can be useful for portfolio management and for investment decisions about hedging climate risk. Alekseev et al. (2022) propose a quantity-based approach to build portfolios that hedge climate change risk. Their approach performs better than traditional methods and can also be used to hedge other risks,

such as aggregate unemployment and house price risk. de Palma and Prigent (2008) conclude that global environmental risks should be hedged with new types of options that combine equity and environmental assets, instead of the current practice of creating separate option markets. Engle et al. (2020) show how to hedge against climate change news. Their study shows that investors should adjust their portfolios to account for climate risks, and that financial strategies should take environmental factors into account.

High temperatures also have important social and policy effects, especially through occupational heat stress. Nunfam et al. (2018) and Nunfam et al. (2019) stress the need to invest in policies against occupational heat stress, with enough financial resources and research. Kjellstrom et al. (2020) describe the economic and social factors that increase health risks in hot workplaces, including gender differences in employment and the vulnerability of migrant workers. Spector and Sheffield (2014) recommend practical and widely applicable methods to assess and control occupational heat stress, in particular in low- and middle-income tropical and subtropical regions.

Given the political debate on climate policy in the US, including the withdrawal from the Paris Agreement, this research can help investment managers and policymakers better understand how investors react to high temperatures.

2 Hypotheses

According to the literature, more extreme heat can harm the economy by reducing labor supply, GDP and exports, especially in sectors that are highly exposed to climate conditions. Pankratz et al. (2023) find that heat lowers firm performance. Addoum et al. (2020) do not find significant effects of high temperatures on sales and labor productivity at the establishment level, even in heat-sensitive industries. However, according to Addoum et al. (2023), the relationship is "bidirectional": high temperatures benefit some industries and harm others. The climate literature looks at temperature shocks in two ways. Some studies examine the effect of temperatures above a certain level, and others the effect of temperatures below a certain level. Studies of the first kind often argue that temperatures above 30°C lower performance, based on evidence such as Pilcher et al. (2002) and Seppänen et al. (2006).

The results of the behavioral papers are also mixed. Cao and Wei (2005) expect lower temperatures to be related to higher returns, while higher temperatures may lead to either higher or lower returns. This is because low temperatures can lead to aggression and risk-taking, while high temperatures can lead to both aggression and apathy. In addition, studies such as Hirshleifer and Shumway (2003) analyze other variables, such as sunshine, which are different from temperature.

Overall, the conclusions of these studies differ and are often contested. As Fisher et al. (2012) note, economists cannot agree on the size, or even the sign, of the effect of climate change on agricultural production.

My research looks at temperature anomalies, that is, deviations from the mean, which are similar to the abnormal temperature used by Ongena and Dursun-de Neef (2024). For this measure, the direction of the effect cannot be predicted in the same way, because the mean temperature differs from place to place and from

season to season. In some places and seasons, a temperature above or below the mean may be good for firms, and in others it may be bad. Addoum et al. (2020) also argue that economic agents adapt to local conditions, so the results may differ between places that are generally hotter or colder. Since many studies point out that the direction of the temperature effect is hard to interpret, I state my hypotheses in terms of the size of the deviation:

1. Larger deviations from the temperature mean affect stock returns (or their standard deviation).
2. Larger deviations from the temperature mean affect stock returns (or their standard deviation) in heat-sensitive industries.

3 Data and methodology

I use a dataset of climate and financial variables to study the effect of temperature anomalies on stock returns and volatility. This section describes the data sources, the variables and the method I use.

3.1 Data description

3.1.1 Financial data

I use Compustat to download the daily stock prices of 18,043 companies that were located and traded in the US between 1995 and 2024. This period is similar to the periods used in related studies, so my results can be compared with theirs. I also download information about each company, such as the city of its headquarters and its exchange code. For industry codes, Compustat offers two classifications, NAICS and GICS. I use NAICS because it is designed for North America and has fewer missing values, which fits the scope of my research.

From the daily stock prices, I calculate the monthly standard deviation of daily log returns as a measure of volatility. I also calculate the monthly log return of each firm, and I use the monthly returns of the S&P 500 index and of the VIX index, in percent, as market controls.

The raw data cover the stock prices of firms across North America. To keep only US firms, I download a list of US cities from Simplemaps.com and remove firms whose headquarters city is in Canada or outside the continental US. I then keep the firms listed on the main US exchanges, which are located in New York, Chicago and Philadelphia.

3.1.2 Climate data

Temperature and precipitation data come from PRISM, the official climatological database of the US Department of Agriculture. PRISM provides daily and monthly maximum, minimum and mean temperatures, as well as precipitation, on grids of 4 km or 800 m.

To download time series from PRISM, I need the latitude and longitude of each location. I use the coordinates of each city from Simplemaps.com, rather than the exact address of each firm. Unlike Pankratz et al. (2023) and Addoum et al. (2020), who use more precise firm locations, I use the climate at the city

Table 1: Summary statistics

	Mean	1st quartile	Median	3rd quartile	SD
Return	−0.003	−0.055	0.001	0.055	0.269
Standard deviation	0.04	0.01	0.02	0.04	0.10
Exchange mean	2.16	0.78	1.74	2.99	1.85
Exchange maximum	7.17	5.39	7.13	8.88	2.87
Exchange minimum	5.98	4.33	6.09	7.53	2.64
Exchange precipitation	1.53	0.50	1.18	2.06	1.44
City mean	2.54	0.91	1.93	3.52	2.27
City maximum	10.26	7.35	10.07	13.00	4.25
City minimum	9.47	6.78	9.26	11.95	4.14
City precipitation	1.49	0.47	1.09	2.03	1.51
Different cities	0.90	1.00	1.00	1.00	0.30
Heat sensitive	0.37	0.00	0.00	1.00	0.48
SP500	0.79	−1.77	1.26	3.60	4.44
VIX	2.25	−12.12	−0.60	10.93	22.70

This table reports summary statistics for the main variables. Return is the monthly log return of the 18,043 US public firms that are located and traded in the continental US. Standard deviation is the monthly standard deviation of these firms' daily returns. The variables starting with "Exchange" and "City" are measured in the city of the exchange and in the city of the headquarters, respectively. Mean, maximum and minimum are the absolute deviations of the monthly mean, maximum and minimum temperatures from the 30-year moving average of the mean temperature in the same calendar month, in degrees Fahrenheit. Precipitation is the absolute deviation of monthly precipitation from its 30-year moving average, in inches. SP500 and VIX are the monthly returns of the two indices, in percent. Different cities is a dummy for firms that are traded in a different city from their headquarters city. Heat sensitive is a dummy for firms in the industries identified as heat-sensitive by Graff Zivin and Neidell (2014).

level. This choice was driven by practical reasons, such as the complexity of the data, and by changing work patterns, in particular remote work after the COVID-19 lockdowns, which make the exact location of a firm less important.

I use these locations to obtain monthly climate records for the sample period. While Addoum et al. (2020) and Pankratz et al. (2023) count the number of days above a certain threshold, I measure the size of the deviation from the long-term average, which gives a different view of how temperature affects risk and return. First, I calculate a 30-year moving average of the monthly mean temperature for each calendar month (for example, over the last 30 Januaries). Then, for the monthly maximum, minimum and mean temperatures in each location, I compute the absolute deviation from this 30-year mean. I calculate the deviation for the maximum and minimum temperatures to capture extreme deviations as well as average ones. The minimum and maximum temperature anomalies can be interpreted as cold and hot extremes relative to the 30-year moving average, respectively. To calculate the precipitation anomaly, I follow the same procedure for monthly precipitation.

Finally, I match each firm's headquarters city with its temperature and precipitation anomalies in each month. This shows how far the temperature of a city in a given month deviates from the historical norm for that month. I also map the climate data to the cities of the exchanges, which gives the temperature and precipitation anomalies for the cities where the firms' stocks are traded.

Table 1 presents summary statistics for the financial and climate variables: the mean, the first quartile, the median, the third quartile and the standard deviation. Temperatures are in degrees Fahrenheit and precipitation is in inches. The labels "mean", "maximum" and "minimum" refer to the deviation of each variable from its 30-year average. For the dummy variables, the mean is the share of observations for which the dummy equals one.

3.2 Methodology

Following the climate economics literature, I estimate the following panel regressions to test the relationship between temperature anomalies and the monthly return or standard deviation:

$$\text{Return}_{i,j,t} = \theta_i + \theta_{j,y} + \beta_1 \text{T}_{i,t} + \beta_2 \text{P}_{i,t} + \beta_3 \text{SP500Return}_t + \epsilon_{i,j,t}$$

$$\text{Standard Deviation}_{i,j,t} = \theta_i + \theta_{j,y} + \beta_1 \text{T}_{i,t} + \beta_2 \text{P}_{i,t} + \beta_3 \text{VIX}_t + \epsilon_{i,j,t}$$

Here, $\text{Return}_{i,j,t}$ is the log return of firm i in industry j in month t, and $\text{Standard Deviation}_{i,j,t}$ is the standard deviation of its daily log returns in that month. $\text{T}_{i,t}$ are the mean, maximum and minimum temperature anomalies in month t in the headquarters and exchange cities of firm i. $\text{P}_{i,t}$ are the precipitation anomalies in the headquarters and exchange cities, which I include as control variables.

I include firm fixed effects (θ_i) to control for time-invariant unobserved differences between firms. Similar

to Pankratz et al. (2023) and Addoum et al. (2020), I include industry-year fixed effects ($\theta_{j,y}$, where y is the year of month t) to control for technological change and other trends within each industry in each year.

I also use two dummy variables in interaction terms:

1. Heat Sensitive is a dummy for heat-sensitive industries, based on the list in Graff Zivin and Neidell (2014) and Addoum et al. (2020). These industries are Agriculture, Forestry, Fishing and Hunting (NAICS codes starting with 11), Mining (21), Utilities (22), Construction (23), Manufacturing (31–33) and Transportation (48–49).
2. Different City is a dummy equal to one if a firm's headquarters is in a different city from the exchange where its stock is traded. As discussed in the introduction, my research is related to two literatures, behavioral finance and climate finance. To combine them, I add the temperature of the exchange city through an interaction with this dummy, which captures the behavioral side of the effect of climate. The interaction avoids using the temperature of the headquarters city twice when the headquarters and the exchange are in the same city.

I do not need to control for the seasonality of temperature, because the temperature anomaly is the deviation from the 30-year average of the same calendar month, so each month is compared with its own long-run mean.

To control for general market movements, I include the S&P 500 return in the return regressions, and the change in the VIX index, as a measure of market volatility, in the standard deviation regressions.

I do not include other firm-level control variables. As Dell et al. (2014) and Angrist and Pischke (2008) point out, controlling for time-varying firm characteristics does not necessarily give a more precise estimate, because these variables may themselves be affected by the main independent variable, the temperature anomaly.

Standard errors are clustered by firm and by month for both return and standard deviation, to account for possible heteroskedasticity and serial correlation in the panel. I follow Addoum et al. (2020), who note that double clustering is more conservative.

4 Main empirical tests and results

4.1 Do temperature anomalies affect monthly returns?

The first question is whether temperature anomalies affect the stock returns of US public companies. Table 2 presents the results of several regressions based on my main model.

In column 1, I regress the firm's log return on the deviation of the monthly mean temperature in the headquarters city from its 30-year moving average (MA). The coefficient is very small and statistically insignificant.

Columns 2 and 3 use the anomalies of the monthly maximum and minimum temperatures, which capture the extreme deviations in each month. Both coefficients are statistically insignificant and not different from zero.

In columns 4, 5 and 6, I estimate the effects of the mean, maximum and minimum temperature anomalies in the cities where the firms' stocks are traded. As for the headquarters cities, these coefficients are statistically insignificant. The t-statistic of the mean temperature anomaly in column 4 (1.17) is larger than the others, but

Table 2: Effect of temperature anomalies on returns

	Return								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Mean Headquarters	9.828×10^{-5}						−0.0003		
	(0.2322)						(−0.8732)		
Max Headquarters		-6.627×10^{-5}						−0.0002	
		(−0.2371)						(−0.8589)	
Min Headquarters			−0.0003						−0.0003
			(−1.0306)						(−1.3291)
Mean Exchange				0.0010					
				(1.1665)					
Max Exchange					0.0002				
					(0.3649)				
Min Exchange						-9.479×10^{-5}			
						(−0.1552)			
Mean Exchange × Different							0.0013		
							(1.4309)		
Max Exchange × Different								0.0004	
								(0.6159)	
Min Exchange × Different									4.254×10^{-5}
									(0.0713)
Precipitation Headquarters	−0.0003	−0.0003	−0.0004				−0.0003	−0.0003	−0.0004
	(−1.2386)	(−1.3064)	(−1.5706)				(−1.5999)	(−1.6492)	(−1.6689)
Precipitation Exchange				−0.0002	−0.0002	−0.0003			
				(−0.2922)	(−0.3014)	(−0.3149)			
Precipitation Exchange × Different							−0.0001	−0.0001	−0.0002
							(−0.1783)	(−0.1454)	(−0.1796)
SP500	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096	0.0096
	(17.839)	(17.949)	(17.924)	(17.385)	(17.659)	(17.753)	(17.456)	(17.697)	(17.774)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0283	0.0283	0.0283	0.0283	0.0283	0.0282	0.0283	0.0283	0.0283
No. observations	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007
No. firms	18,043	18,043	18,043	18,043	18,043	18,043	18,043	18,043	18,043

This table reports the effect of different temperature anomaly measures on the returns of US firms. The dependent variable is the monthly log return. The temperature variables are the absolute deviations of the monthly mean, maximum and minimum temperatures from the 30-year moving average of the same calendar month. Columns 7 to 9 add interaction terms between the exchange-city variables and Different. Different is a dummy for firms whose exchange is in a different city from their headquarters. t-statistics based on standard errors clustered by firm and by month are reported in parentheses below the coefficients. Significance stars are shown for the temperature anomaly variables: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

still far from significant. The coefficient means that a 1 degree Fahrenheit increase in the size of the deviation of the monthly mean temperature from its 30-year MA increases the monthly log return by 0.0010, or about 0.1 percent.

Columns 7, 8 and 9 look at another part of my research question. They include the temperature anomalies of the headquarters and of the exchange city in the same regression, using the interaction with the dummy for firms whose headquarters and exchange are in different cities. None of the temperature coefficients is significant at the 1, 5 or 10 percent level. These results suggest that long-run temperature deviations may not be a good measure of investor mood, and that daily weather variables, such as sunshine, cloud cover or daily temperature, may be better predictors than climate measures.

Overall, none of the temperature anomaly coefficients is statistically different from zero. Temperature anomalies at the headquarters or at the exchange do not affect monthly returns. These results are in line with the climate economics studies that find no significant relationship between temperature and economic outcomes in the US, such as Addoum et al. (2020).

4.2 Do temperature anomalies affect the monthly standard deviation?

In this section, I examine whether temperature anomalies affect the standard deviation of the stock returns of US public companies. Table 3 presents the results of several regressions based on the main model.

In column 1, I regress the monthly standard deviation of the firm's returns on the deviation of the monthly mean temperature in the headquarters city from its 30-year MA. The coefficient is not statistically significant, but its t-statistic of 1.60 is close to the 10 percent level. It shows that a 1 degree Fahrenheit increase in the mean temperature anomaly at the headquarters increases the standard deviation of the firm's returns by 0.0017.

The coefficient of the maximum temperature anomaly in column 2 is statistically insignificant.

The coefficient of the minimum temperature anomaly at the headquarters in column 3 is significant at the 10 percent level. If the deviation of the monthly minimum temperature from its 30-year average at the headquarters increases by 1 degree Fahrenheit, the standard deviation of the firm's returns decreases by 0.0011.

Columns 4 to 6 estimate the effect of temperature in the exchange city on volatility, by regressing the monthly standard deviation on the mean, maximum and minimum temperature anomalies. The coefficient of the mean temperature anomaly at the exchange in column 4 is significant at the 10 percent level: a 1 degree Fahrenheit increase in the absolute deviation of the monthly mean temperature from its long-run average increases the standard deviation by 0.0032. The coefficient of the maximum temperature anomaly at the exchange in column 5 is not significantly different from zero.

In column 6, the coefficient shows that a 1 degree Fahrenheit increase in the minimum temperature anomaly at the exchange reduces the standard deviation by 0.0021. The coefficient is significant at the 10 percent level, and the result is similar to that for the minimum temperature anomaly at the headquarters in column 3.

In column 7, I use the interaction term to include the headquarters and the exchange temperature anomalies in the same regression. The coefficient of the interaction term (Mean Exchange $\times$ Different) is significant at

Table 3: Effect of temperature anomalies on standard deviation

	Standard deviation								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Mean Headquarters	0.0017						0.0009		
	(1.6002)						(1.4356)		
Max Headquarters		0.0005						4.611×10^{-5}	
		(1.1031)						(0.1734)	
Min Headquarters			−0.0011*						−0.0006*
			(−1.9084)						(−1.7275)
Mean Exchange				0.0032*					
				(1.6888)					
Max Exchange					0.0014				
					(1.5328)				
Min Exchange						−0.0021*			
						(−1.8100)			
Mean Exchange × Different							0.0028*		
							(1.6769)		
Max Exchange × Different								0.0014	
								(1.5924)	
Min Exchange × Different									−0.0018*
									(−1.7136)
Precipitation Headquarters	0.0002	0.0001	−0.0001				9.652×10^{-5}	1.238×10^{-5}	−0.0001
	(1.3045)	(0.9900)	(−1.0380)				(0.8136)	(0.1256)	(−0.8428)
Precipitation Exchange				−0.0003	−0.0003	−0.0005			
				(−0.4866)	(−0.5157)	(−0.8433)			
Precipitation Exchange × Different							−0.0002	−0.0003	−0.0005
							(−0.4018)	(−0.5068)	(−0.8111)
VIX	-7.155×10^{-6}	-1.849×10^{-5}	-2.039×10^{-5}	1.791×10^{-6}	-1.631×10^{-5}	-2.737×10^{-5}	3.025×10^{-6}	-1.669×10^{-5}	-2.6×10^{-5}
	(−0.2015)	(−0.5482)	(−0.5952)	(0.0464)	(−0.4656)	(−0.7238)	(0.0779)	(−0.4802)	(−0.6970)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0019	0.0004	0.0018	0.0045	0.0021	0.0038	0.0046	0.0019	0.0042
No. observations	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007
No. firms	18,043	18,043	18,043	18,043	18,043	18,043	18,043	18,043	18,043

This table reports the effect of different temperature anomaly measures on the standard deviation of the returns of US firms. The dependent variable is the monthly standard deviation of daily log returns. The temperature variables are the absolute deviations of the monthly mean, maximum and minimum temperatures from the 30-year moving average of the same calendar month. Columns 7 to 9 add interaction terms between the exchange-city variables and Different. Different is a dummy for firms whose exchange is in a different city from their headquarters. t-statistics based on standard errors clustered by firm and by month are reported in parentheses below the coefficients. Significance stars are shown for the temperature anomaly variables: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the 10 percent level, which confirms the result in column 4. For companies whose headquarters are not in the same city as their exchange, a 1 degree increase in the mean temperature anomaly in the exchange city increases the standard deviation by 0.0028.

In column 8, the coefficients of the maximum temperature anomaly at the headquarters and of the interaction term (Max Exchange × Different) are not significantly different from zero.

In column 9, both the minimum temperature anomaly at the headquarters and the interaction term (Min Exchange × Different) are significant at the 10 percent level. A 1 degree Fahrenheit increase in the size of the deviation of the monthly minimum temperature from its 30-year average decreases the standard deviation by 0.0006 at the headquarters and by 0.0018 at the exchange.

The coefficients of the mean temperature anomaly at the headquarters and at the exchange (columns 1 and 4) and of the interaction term (Mean Exchange × Different) in column 7 are all positive, although the coefficient in column 1 is not significant. Since the mean temperature anomaly is measured in absolute terms, it does not show whether a month was warmer or colder than normal. The positive sign therefore means that larger deviations of the monthly mean temperature from its long-run average, in either direction, increase the standard deviation. Based on the literature discussed above, one explanation is that these deviations are shocks to which firms and people have not adapted.

In contrast, the coefficients of the minimum temperature anomaly at the headquarters, at the exchange and of the interaction term (Min Exchange × Different) are negative. Since the minimum temperature anomaly captures cold extremes, this suggests that the standard deviation decreases when it becomes colder relative to the 30-year monthly average. This is in line with Tzouvanas et al. (2019), who find that cold shocks reduce systemic risk.

Overall, the t-statistics in the standard deviation regressions are higher than those in the return regressions. This suggests that deviations of the monthly mean, maximum and minimum temperatures from their long-run averages are more closely related to volatility than to returns. In addition, the headquarters and the exchange show similar results. This suggests that the standard deviation of US stocks may be sensitive to temperature variability both through investor sentiment and through the effect of climate risk on firms' infrastructure.

4.3 Heat-sensitive industries

To isolate the effect of temperature anomalies on the heat-sensitive industries identified by Graff Zivin and Neidell (2014), I interact the heat-sensitive dummy with the climate variables. I run separate regressions for the mean, maximum and minimum anomalies in both the headquarters and the exchange cities. The idea is to better understand how temperature affects firm performance and investor sentiment, and through them the stock market, in these specific industries.

4.3.1 Return

Panel A of Table 4 shows the results for the returns of US firms in these industries. In column 1, I do not find significant results for the mean temperature anomaly in the headquarters city. In columns 2 and 3, which

Table 4: Effect of temperature anomalies on the returns and standard deviation of heat-sensitive industries

	A. Return						B. Standard deviation					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Mean Headquarters	-2.04×10^{-5}						0.0015					
	(−0.0516)						(1.5674)					
× Heat Sensitive	0.0003						0.0003**					
	(1.0778)						(1.9990)					
Max Headquarters		−0.0001						0.0005				
		(−0.4115)						(1.1566)				
× Heat Sensitive		0.0001						1.489×10^{-5}				
		(0.6397)						(0.2043)				
Min Headquarters			−0.0002						−0.0010*			
			(−0.6563)						(−1.8471)			
× Heat Sensitive			−0.0004						−0.0003**			
			(−1.4122)						(−2.0509)			
Mean Exchange				0.0007						0.0028*		
				(0.9121)						(1.6872)		
× Heat Sensitive				0.0009						0.0008		
				(1.6017)						(1.6240)		
Max Exchange					0.0002						0.0013	
					(0.3061)						(1.5499)	
× Heat Sensitive					0.0002						0.0004	
					(0.4221)						(1.4376)	
Min Exchange						−0.0002						−0.0019*
						(−0.4310)						(−1.8028)
× Heat Sensitive						0.0003						−0.0006
						(0.7749)						(−1.6360)
Precipitation Headquarters	−0.0004	−0.0004	−0.0004				0.0002	0.0001	-7.753×10^{-5}			
	(−1.5740)	(−1.6119)	(−1.7040)				(1.3010)	(1.0670)	(−0.6436)			
× Heat Sensitive	0.0002	0.0001	4.925×10^{-5}				3.952×10^{-5}	1.23×10^{-5}	−0.0001			
	(0.4647)	(0.4336)	(0.1455)				(0.3331)	(0.1027)	(−0.8530)			
Precipitation Exchange				−0.0004	−0.0004	−0.0004				−0.0002	−0.0002	−0.0004
				(−0.5690)	(−0.5789)	(−0.6072)				(−0.3629)	(−0.4047)	(−0.7530)
× Heat Sensitive				0.0004	0.0004	0.0004				−0.0002	−0.0002	−0.0003
				(0.6179)	(0.6056)	(0.6675)				(−0.8584)	(−0.8153)	(−0.9862)
SP500 (A) / VIX (B)	0.0087	0.0087	0.0087	0.0086	0.0086	0.0086	-5.157×10^{-6}	-1.522×10^{-5}	-1.746×10^{-5}	2.601×10^{-6}	-1.377×10^{-5}	-2.401×10^{-5}
	(16.504)	(16.623)	(16.619)	(15.994)	(16.295)	(16.399)	(−0.1587)	(−0.4912)	(−0.5539)	(0.0733)	(−0.4262)	(−0.6860)
× Heat Sensitive	0.0026	0.0026	0.0026	0.0026	0.0026	0.0026	1.368×10^{-6}	-1.636×10^{-6}	-8.897×10^{-7}	3.929×10^{-6}	-5.766×10^{-7}	-3.135×10^{-6}
	(8.1685)	(8.1651)	(8.1701)	(8.0824)	(8.1551)	(8.2189)	(0.1206)	(−0.1467)	(−0.0788)	(0.3607)	(−0.0524)	(−0.2625)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0287	0.0287	0.0288	0.0288	0.0287	0.0287	0.0019	0.0004	0.0020	0.0048	0.0022	0.0042
No. observations	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007
No. firms	18,043	18,043	18,043	18,043	18,043	18,043	18,043	18,043	18,043	18,043	18,043	18,043

This table reports the effect of different temperature anomaly measures on firms in heat-sensitive industries: Agriculture, Forestry, Fishing and Hunting (NAICS codes starting with 11), Mining (21), Utilities (22), Construction (23), Manufacturing (31–33) and Transportation (48–49). The dependent variable is the monthly log return in Panel A and the monthly standard deviation in Panel B. The temperature variables are the absolute deviations of the monthly mean, maximum and minimum temperatures from the 30-year moving average of the same calendar month. Each column includes an interaction term between these measures and the heat-sensitive dummy. The row SP500 (A) / VIX (B) shows the S&P 500 return in Panel A and the change in the VIX in Panel B. t-statistics based on standard errors clustered by firm and by month are reported in parentheses below the coefficients. Significance stars are shown for the temperature anomaly variables: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

use the maximum and minimum temperature anomalies, the coefficients for heat-sensitive industries are not different from zero.

In columns 4, 5 and 6, the climate variables of the exchange cities do not give significant results either. The only notable result is the interaction term between the mean temperature anomaly at the exchange and the heat-sensitive dummy, with a t-statistic of 1.60. It shows that for firms in heat-sensitive industries, a 1 degree Fahrenheit increase in the size of the deviation of the monthly mean temperature from its 30-year moving average increases the return by 0.0009 (about 0.09 percent) more than for other firms. This difference is small and not statistically significant.

In general, my hypothesis that the absolute value of the monthly temperature anomaly significantly affects the returns of US firms in heat-sensitive industries is not supported by the evidence.

4.3.2 Standard deviation

Panel B of Table 4 shows the effect of temperature anomalies on the monthly standard deviation of the stock returns of firms in heat-sensitive industries. In column 7, the interaction term with the heat-sensitive dummy is significant at the 5 percent level. For firms in heat-sensitive industries, a 1 degree Fahrenheit increase in the mean temperature anomaly at the headquarters increases the monthly standard deviation by 0.0003 more than for other firms.

In columns 8 and 11, which use the maximum temperature anomaly at the headquarters and at the exchange, none of the temperature variables or their interaction terms is significant.

In column 9, the coefficient of the minimum temperature anomaly at the headquarters and its interaction with the heat-sensitive dummy are significant at the 10 and 5 percent levels, respectively. A 1 degree Fahrenheit increase in the minimum temperature anomaly decreases the standard deviation by 0.0010 for firms in other industries, and by a further 0.0003 (0.0013 in total) for firms in heat-sensitive industries.

For the exchanges, the notable results are in columns 10 and 12. The mean and minimum temperature anomalies are significant at the 10 percent level for firms in non-heat-sensitive industries, but their interaction terms with the heat-sensitive dummy are not significant. Column 10 shows that a 1 degree Fahrenheit increase in the mean temperature anomaly at the exchange increases the standard deviation of firms in non-heat-sensitive industries by 0.0028. In column 12, a 1 degree Fahrenheit increase in the minimum temperature anomaly at the exchange decreases their standard deviation by 0.0019. In both cases, the results suggest that temperature shocks in exchange cities affect investor mood regardless of the industry.

At the headquarters, however, the results are stronger for heat-sensitive industries than for other industries. This means that temperature anomalies at the headquarters have an additional effect on the standard deviation of firms in heat-sensitive industries. In other words, the stock volatility of these firms seems to be sensitive to temperature anomalies through their headquarters and infrastructure.

As in Table 3, the minimum temperature anomaly has negative coefficients and the mean temperature anomaly has positive coefficients, while the maximum temperature anomaly is not significant. In addition, the heat-sensitive interaction terms in columns 7 and 9 are significant at the 5 percent level, which is stronger than the coefficients for the other firms. This shows that the stock volatility of firms in heat-sensitive industries is

more sensitive to temperature anomalies.

In summary, my hypothesis that the absolute value of the monthly temperature anomaly significantly affects the standard deviation of US firms in heat-sensitive industries is partly supported by the evidence.

5 Robustness checks

In this section, I run three additional tests to check the robustness of the baseline model. First, I analyze whether lagged temperature anomalies affect returns and volatility. Second, I test whether firms in warmer and colder areas react differently to temperature anomalies. Third, I replace the industry-year fixed effects with industry-quarter fixed effects to better capture variation within a year.

5.1 Lagged temperature effect

To check the robustness of my main results, I add one-quarter lags of the temperature anomaly measures. This lets me examine both the immediate and the lagged effects of temperature anomalies on returns and standard deviation.

According to Pankratz et al. (2023), high temperatures can lower firm performance one to two quarters after exposure. Barrot and Sauvagnat (2016) show that shocks from natural disasters affect firm performance both immediately and in later periods, so lagged variables are important to capture the full effect of such shocks.

The literature suggests that the market may not react immediately to temperature anomalies. Instead, it adjusts as firms publish their financial results and more information becomes available. These results are published every quarter and show the possible effects of temperature shocks. As Pankratz et al. (2023) show, analysts may then be surprised by the announcements and update their forecasts.

According to Daniel et al. (1998), investors may underreact to new information because of cognitive biases, which delays the market response. The market may therefore not react fully to a temperature anomaly at first, but react more as its effects become clear over time. Such shocks often lead analysts and investors to adjust stock prices in the following quarters. For this reason, it is important to look at both the immediate and the lagged effects of temperature anomalies on stock returns and standard deviation.

Table A1 in the Appendix shows no significant results, in line with the main results, except for the lag of the minimum temperature anomaly in column 6. This coefficient is significant at the 5 percent level. It shows that a 1 degree Fahrenheit increase in the deviation of the monthly minimum temperature from its 30-year MA is associated with a higher return three months later, by 0.0005 (about 0.05 percent).

According to this result, a larger deviation of the monthly minimum temperature from its long-run average is likely to increase the stock return in the following quarter.

Generally, the results of this test indicate that my main analysis is robust, and that monthly returns are not significantly related to the one-quarter lags of the temperature anomalies.

Table A2 in the Appendix shows the results of regressing the monthly standard deviation of US firms' stock returns on the temperature anomaly variables and their one-quarter lags.

Two temperature coefficients are significant. The mean temperature anomaly at the headquarters in column

4 is significant at the 10 percent level, and a 1 degree Fahrenheit increase in this anomaly increases the standard deviation by 0.0008. In addition, the one-quarter lag of the mean temperature anomaly at the exchange (Mean Exchange × Different) is significant at the 5 percent level. It shows that a 1 degree Fahrenheit increase reduces the standard deviation three months later by 0.0008. Both coefficients, however, are economically very small.

Compared with Table 3, the results of this analysis are less significant. This suggests that deviations in monthly temperature are more likely to affect the standard deviation immediately than with a delay, although these effects are also economically very small.

Overall, apart from a few weak results, I find no one-quarter lagged effect of the monthly temperature anomalies.

5.2 Adaptation in warmer locations

According to Kotz et al. (2021), increases in day-to-day temperature variability, measured as the standard deviation of daily temperatures within a month, reduce economic growth. The losses are largest in low-latitude regions, which have small seasonal temperature differences and are less resilient to daily temperature fluctuations. Warmer locations may therefore be more sensitive to temperature anomalies within a month than colder locations.

As in Addoum et al. (2020), I split the sample into warmer and colder locations. Addoum et al. (2020) split locations based on their annual average temperatures, while I use the 100-year average temperature of each US city from 1920 to 2020 and define warmer locations as those above the median. I test this idea with interaction terms between the warmer-location dummy and the different temperature anomaly measures in both the headquarters and the exchange cities, for both returns and standard deviation.

In Table A3, I regress the log return of US firms on the mean, maximum and minimum temperature anomalies at the headquarters and at the exchanges, and on their interactions with the dummy for warmer locations. None of the interaction terms is statistically significant, so firms located in warmer climates do not react differently. The largest t-statistic is that of (Min Headquarters × Warmer), at 1.23, which is still not significantly different from zero.

For the monthly standard deviation, these tests give results similar to the baseline regressions. None of the interaction terms with the warmer-location dummy in Table A4 in the Appendix is significant. The interactions between the warmer dummy and the headquarters temperature anomalies have larger t-statistics than those with the exchange temperature anomalies, especially for the maximum temperature, which may point to an effect of high temperatures. However, they are not different from zero. The coefficients and t-statistics of the interactions between the warmer dummy and the exchange temperature anomalies are very small.

At most, this hints that the standard deviation of US stocks may be more sensitive to temperature anomalies at the headquarters than at the exchanges. One reason could be that investors at the exchanges do not care whether a headquarters is in a warmer region, since they are not involved in its operations.

Overall, I do not find significant evidence that firms in warmer cities have different monthly returns or standard deviations from firms in colder cities.

5.3 Industry-quarter fixed effects

Within a year, each industry is subject to seasonality and short-term economic cycles, and some industries have their own seasons, such as the harvest in agriculture. This variation may affect my interpretation of the effect of temperature anomalies in many industries. To capture unobserved differences that vary by quarter and industry, I use industry-quarter fixed effects. I regress the log return and the standard deviation of the firms in my sample on the temperature anomalies at the headquarters, and on the interactions of the exchange temperature variables with the Different City dummy for firms whose headquarters and exchange are in different cities.

To avoid confounding, I do not include the heat-sensitive interaction terms in the regressions with industry-quarter fixed effects. Heat-sensitive industries may have their own temperature patterns, such as consistently higher temperatures in certain seasons, so the industry-quarter fixed effects may already absorb part of the effects I would like to measure with the heat-sensitive interaction terms.

Table A5 shows the effect of the temperature anomalies on the log returns of US firms with industry-quarter fixed effects. In general, the results are similar to the main analysis. However, the coefficients of the minimum temperature anomaly at the headquarters in columns 3 and 6 are now significant, at the 5 and 10 percent levels, respectively. This suggests that the main model may not capture some seasonal effects. Although these coefficients are significant, they are small, and their signs are the same as in the regressions with industry-year fixed effects.

For returns, the results of this alternative specification are therefore consistent with my main model. The main results and conclusions are robust, but they should be interpreted with caution because of the variation within industries over the year.

Table A6 shows the effect of the temperature anomalies on the monthly standard deviation of US firms with industry-quarter fixed effects. The results show an interesting pattern. Compared with the industry-year fixed effects (Table 3, columns 7 to 9), the t-statistics of the exchange temperature variables in columns 4, 5 and 6 clearly increase, and the interaction term of the maximum temperature anomaly at the exchange becomes significant at the 10 percent level, while it was not significant before. In column 3, the minimum temperature anomaly at the headquarters is significant at the 5 percent level, but it is no longer significant once the interaction term of the minimum exchange temperature anomaly is added in column 6.

This suggests that the seasonality of each industry is more closely related to the headquarters temperature than to the exchange temperature. When I control for this seasonality with industry-quarter fixed effects, the headquarters temperature anomalies become weaker and are not significant in the regressions that also include the exchange variables (columns 4 to 6).

The results suggest that the effect of temperature at the headquarters is seasonal and depends on the quarter, while temperature in the exchange cities can affect investor sentiment regardless of the seasonality of the industry.

Both specifications are robust, but they differ in how strongly the headquarters and the exchange temperatures matter. Which one is more suitable depends on the aim of the analysis: either to accept that each industry has its own seasonal pattern, or to control for these patterns and compare all months on an equal basis, leaving

out the extreme effects that can occur in particular quarters.

6 Conclusion

My research examines how the deviations of the mean, maximum and minimum monthly temperatures from the 30-year moving average of the same month affect the return and risk of US stocks. I study the possible effect of temperature anomalies through two channels: the city of the headquarters, where temperature can affect firms' infrastructure and workers, and the city of the exchange, where it can affect investor mood. I also repeat this analysis for heat-sensitive industries.

In most of my tests, the results are consistent with those of Addoum et al. (2020). The relationship between temperature anomalies and returns is economically and statistically insignificant for both the headquarters and the exchange. This also holds for the returns of firms in heat-sensitive industries. Only a few coefficients in the robustness checks are significant, and they are small.

However, I find that some measures of temperature anomalies affect the monthly standard deviation of the stock returns of US public companies, although most of these effects are significant only at the 10 percent level. Deviations of the monthly minimum temperature from its 30-year mean are negatively related to the standard deviation, at both the headquarters and the exchange. This suggests that cold shocks may reduce stock market volatility. In addition, deviations of the monthly mean temperature from its long-run average, which I see as an indicator of how well firms and people are adapted to their local temperature, are positively related to the standard deviation, because they bring more variation in temperature.

Interacting the temperature anomalies with the heat-sensitive dummy gives another result. Temperature shocks in exchange cities seem to affect investor mood regardless of the industry. At the headquarters, however, the results are stronger for heat-sensitive industries than for other industries. This suggests that the stock volatility of firms in heat-sensitive industries is sensitive to temperature anomalies through their headquarters and infrastructure, while at the exchange it makes no difference whether a firm is in a heat-sensitive industry or not.

Finally, this study can serve as a starting point for further research. The idea could be developed with more precise methods such as geocoding, which gives the exact coordinates of all of a firm's properties, not only the headquarters but also its branches and plants. Another new research question is how the spread of a firm's branches across warm and cold locations affects the resilience of its stock and profitability to changes in temperature.

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A Appendix

Table A1: Lagged temperature effect: returns

	Return					
	(1)	(2)	(3)	(4)	(5)	(6)
Mean Headquarters	7.091×10^{-5}			−0.0003		
	(0.1650)			(−1.0859)		
Mean Headquarters($t-3$)	0.0003			0.0003		
	(1.0259)			(1.1377)		
Max Headquarters		-8.084×10^{-5}			−0.0002	
		(−0.2868)			(−1.0256)	
Max Headquarters($t-3$)		-2.537×10^{-5}			-1.136×10^{-5}	
		(−0.1120)			(−0.0573)	
Min Headquarters			−0.0003			−0.0003
			(−0.9744)			(−1.2778)
Min Headquarters($t-3$)			0.0004			0.0005**
			(1.3887)			(2.0453)
Mean Exchange × Different				0.0013		
				(1.4304)		
Mean Exchange × Different($t-3$)				0.0003		
				(0.3763)		
Max Exchange × Different					0.0004	
					(0.6069)	
Max Exchange × Different($t-3$)					-5.29×10^{-5}	
					(−0.0973)	
Min Exchange × Different						0.0001
						(0.2042)
Min Exchange × Different($t-3$)						−0.0004
						(−0.6954)
Precipitation Headquarters	−0.0003	−0.0003	−0.0004	−0.0004	−0.0004	−0.0004
	(−1.2993)	(−1.3504)	(−1.6712)	(−1.6636)	(−1.7043)	(−1.6515)
Precipitation Headquarters($t-3$)	0.0002	0.0002	0.0002	0.0001	0.0001	0.0002
	(0.7813)	(0.6577)	(0.9268)	(0.5663)	(0.4989)	(0.7922)
Precipitation Exchange × Different				−0.0001	-7.165×10^{-5}	−0.0001
				(−0.1264)	(−0.0846)	(−0.1235)
Precipitation Exchange × Different($t-3$)				0.0009	0.0009	0.0009
				(0.8586)	(0.8456)	(0.8196)
SP500	0.0096	0.0096	0.0097	0.0096	0.0096	0.0096
	(17.927)	(18.003)	(18.041)	(17.515)	(17.805)	(17.929)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0289	0.0289	0.0290	0.0290	0.0290	0.0290
No. observations	1,909,202	1,909,202	1,909,202	1,909,202	1,909,202	1,909,202
No. firms	17,749	17,749	17,749	17,749	17,749	17,749

This table reports the effect of temperature anomalies and their one-quarter lags ($t-3$) on the monthly log return of US public firms. The temperature variables are the absolute deviations of the monthly mean, maximum and minimum temperatures from the 30-year moving average of the same calendar month. Columns 4 to 6 include interaction terms between the exchange-city variables and Different. Different is a dummy for firms whose exchange is in a different city from their headquarters. t-statistics based on standard errors clustered by firm and by month are reported in parentheses below the coefficients. Significance stars are shown for the temperature anomaly variables: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A2: Lagged temperature effect: standard deviation

	Standard deviation					
	(1)	(2)	(3)	(4)	(5)	(6)
Mean Headquarters	0.0015			0.0008*		
	(1.4465)			(1.6872)		
Mean Headquarters$(t-3)$	−0.0002			0.0002		
	(−1.0990)			(0.8934)		
Max Headquarters		0.0004			8.395×10^{-5}	
		(1.0339)			(0.4248)	
Max Headquarters$(t-3)$		0.0002			0.0003	
		(1.4046)			(1.6347)	
Min Headquarters			−0.0008			−0.0005
			(−1.5538)			(−1.6114)
Min Headquarters$(t-3)$			-7.423×10^{-5}			3.057×10^{-5}
			(−0.7435)			(0.2415)
Mean Exchange × Different				0.0019		
				(1.4203)		
Mean Exchange × Different$(t-3)$				−0.0008**		
				(−1.9682)		
Max Exchange × Different					0.0009	
					(1.3099)	
Max Exchange × Different$(t-3)$					−0.0003	
					(−1.1431)	
Min Exchange × Different						−0.0007
						(−1.1077)
Min Exchange × Different$(t-3)$						−0.0003
						(−1.1330)
Precipitation Headquarters	0.0001	7.019×10^{-5}	−0.0001	3.842×10^{-5}	-3.834×10^{-5}	-8.922×10^{-5}
	(1.0281)	(0.6663)	(−1.0358)	(0.4243)	(−0.4669)	(−0.9797)
Precipitation Headquarters$(t-3)$	0.0010	0.0011	0.0010	0.0008	0.0009	0.0008
	(1.5807)	(1.5778)	(1.5206)	(1.7475)	(1.7088)	(1.6280)
Precipitation Exchange × Different				0.0003	0.0003	0.0002
				(0.6997)	(0.6349)	(0.4049)
Precipitation Exchange × Different$(t-3)$				0.0038	0.0037	0.0036
				(1.3123)	(1.2691)	(1.2135)
VIX	9.02×10^{-6}	6.672×10^{-7}	-9.589×10^{-7}	2.954×10^{-6}	-8.055×10^{-6}	-9.932×10^{-6}
	(0.2879)	(0.0229)	(−0.0323)	(0.0936)	(−0.2747)	(−0.3209)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0023	0.0010	0.0016	0.0088	0.0064	0.0066
No. observations	1,909,202	1,909,202	1,909,202	1,909,202	1,909,202	1,909,202
No. firms	17,749	17,749	17,749	17,749	17,749	17,749

This table reports the effect of temperature anomalies and their one-quarter lags $(t-3)$ on the monthly standard deviation of the daily returns of US public firms. The temperature variables are the absolute deviations of the monthly mean, maximum and minimum temperatures from the 30-year moving average of the same calendar month. Columns 4 to 6 include interaction terms between the exchange-city variables and Different. Different is a dummy for firms whose exchange is in a different city from their headquarters. t-statistics based on standard errors clustered by firm and by month are reported in parentheses below the coefficients. Significance stars are shown for the temperature anomaly variables: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Accounting for adaptation: returns

	Return					
	(1)	(2)	(3)	(4)	(5)	(6)
Mean Headquarters	0.0003					
	(0.7222)					
× Warmer	−0.0004					
	(−0.8709)					
Max Headquarters		-9.836×10^{-5}				
		(−0.3584)				
× Warmer		6.064×10^{-5}				
		(0.2512)				
Min Headquarters			−0.0005			
			(−1.5880)			
× Warmer			0.0003			
			(1.2329)			
Mean Exchange				0.0010		
				(1.1658)		
× Warmer				1.52×10^{-5}		
				(0.0643)		
Max Exchange					0.0003	
					(0.4632)	
× Warmer					−0.0001	
					(−0.6665)	
Min Exchange						−0.0001
						(−0.1693)
× Warmer						1.668×10^{-5}
						(0.0971)
Precipitation Headquarters	−0.0001	−0.0002	−0.0003			
	(−0.2980)	(−0.4517)	(−0.8036)			
× Warmer	−0.0003	−0.0002	-8.791×10^{-5}			
	(−0.5680)	(−0.4483)	(−0.1925)			
Precipitation Exchange				−0.0002	−0.0002	−0.0002
				(−0.2241)	(−0.2280)	(−0.2411)
× Warmer				-8.419×10^{-5}	-9.133×10^{-5}	-9.228×10^{-5}
				(−0.2793)	(−0.3056)	(−0.3099)
SP500	0.0096	0.0096	0.0096	0.0095	0.0096	0.0096
	(17.768)	(17.856)	(17.784)	(17.350)	(17.570)	(17.664)
× Warmer	7.921×10^{-5}	6.529×10^{-5}	7.663×10^{-5}	6.501×10^{-5}	6.857×10^{-5}	6.744×10^{-5}
	(0.5245)	(0.4380)	(0.5118)	(0.4354)	(0.4602)	(0.4522)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0283	0.0283	0.0283	0.0283	0.0283	0.0282
No. observations	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007
No. firms	18,043	18,043	18,043	18,043	18,043	18,043

This table reports the effect of temperature anomalies for firms in warmer locations, defined as cities with a 100-year average temperature above the median of US cities. The dependent variable is the monthly log return. The temperature variables are the absolute deviations of the monthly mean, maximum and minimum temperatures from the 30-year moving average of the same calendar month. Each column includes an interaction term between these measures and the warmer-location dummy. t-statistics based on standard errors clustered by firm and by month are reported in parentheses below the coefficients. Significance stars are shown for the temperature anomaly variables: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Accounting for adaptation: standard deviation

	Standard deviation					
	(1)	(2)	(3)	(4)	(5)	(6)
Mean Headquarters	0.0015					
	(1.4038)					
× Warmer	0.0003					
	(1.1966)					
Max Headquarters		0.0003				
		(0.8724)				
× Warmer		0.0003				
		(1.3774)				
Min Headquarters			−0.0011**			
			(−1.9769)			
× Warmer			-3.467×10^{-5}			
			(−0.1897)			
Mean Exchange				0.0032*		
				(1.6953)		
× Warmer				-3.921×10^{-5}		
				(−0.3722)		
Max Exchange					0.0014	
					(1.5776)	
× Warmer					-4.137×10^{-5}	
					(−0.7264)	
Min Exchange						−0.0021*
						(−1.8182)
× Warmer						4.083×10^{-5}
						(0.4282)
Precipitation Headquarters	0.0006	0.0004	1.731×10^{-5}			
	(1.0148)	(0.7780)	(0.0443)			
× Warmer	−0.0005	−0.0003	−0.0002			
	(−0.6966)	(−0.4581)	(−0.3091)			
Precipitation Exchange				−0.0003	−0.0003	−0.0005
				(−0.4250)	(−0.4338)	(−0.7566)
× Warmer				-4.909×10^{-6}	-2.817×10^{-5}	-3.073×10^{-5}
				(−0.0563)	(−0.3428)	(−0.3615)
VIX	-7.899×10^{-7}	-1.286×10^{-5}	-1.2×10^{-5}	6.887×10^{-6}	-1.132×10^{-5}	-2.234×10^{-5}
	(−0.0217)	(−0.3837)	(−0.3537)	(0.1800)	(−0.3232)	(−0.5875)
× Warmer	-6.159×10^{-6}	-4.797×10^{-6}	-9.572×10^{-6}	-4.535×10^{-6}	-4.3×10^{-6}	-4.669×10^{-6}
	(−0.9849)	(−0.9500)	(−1.4974)	(−0.9843)	(−0.9598)	(−0.9945)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry-year FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0019	0.0004	0.0020	0.0048	0.0022	0.0041
No. observations	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007
No. firms	18,043	18,043	18,043	18,043	18,043	18,043

This table reports the effect of temperature anomalies on the standard deviation of firms in warmer locations, defined as cities with a 100-year average temperature above the median of US cities. The dependent variable is the monthly standard deviation. The temperature variables are the absolute deviations of the monthly mean, maximum and minimum temperatures from the 30-year moving average of the same calendar month. Each column includes an interaction term between these measures and the warmer-location dummy. t-statistics based on standard errors clustered by firm and by month are reported in parentheses below the coefficients. Significance stars are shown for the temperature anomaly variables: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Effect of temperature anomalies on returns (industry-quarter fixed effects)

	Return					
	(1)	(2)	(3)	(4)	(5)	(6)
Mean Headquarters	0.0002			-7.792×10^{-5}		
	(0.6689)			(−0.3145)		
Max Headquarters		3.108×10^{-5}			-4.803×10^{-5}	
		(0.1453)			(−0.2916)	
Min Headquarters			−0.0005**			−0.0004*
			(−2.3462)			(−1.8781)
Mean Exchange × Different				0.0015		
				(1.6111)		
Max Exchange × Different					0.0003	
					(0.6037)	
Min Exchange × Different						−0.0007
						(−1.1555)
Precipitation Headquarters	-6.607×10^{-6}	-1.616×10^{-5}	−0.0001	-6.118×10^{-5}	-4.744×10^{-5}	−0.0001
	(−0.0324)	(−0.0784)	(−0.5889)	(−0.3249)	(−0.2529)	(−0.6486)
Precipitation Exchange × Different				8.28×10^{-5}	0.0001	-1.328×10^{-5}
				(0.0940)	(0.1512)	(−0.0151)
SP500	0.0088	0.0088	0.0088	0.0087	0.0088	0.0088
	(17.407)	(17.427)	(17.469)	(17.371)	(17.495)	(17.342)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0171	0.0171	0.0171	0.0172	0.0171	0.0172
No. observations	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007
No. firms	18,043	18,043	18,043	18,043	18,043	18,043

This table reports the effect of different temperature anomaly measures on the returns of US firms with industry-quarter fixed effects. The dependent variable is the monthly log return. The temperature variables are the absolute deviations of the monthly mean, maximum and minimum temperatures from the 30-year moving average of the same calendar month. Columns 4 to 6 include interaction terms between the exchange-city variables and Different. Different is a dummy for firms whose exchange is in a different city from their headquarters. t-statistics based on standard errors clustered by firm and by month are reported in parentheses below the coefficients. Significance stars are shown for the temperature anomaly variables: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Effect of temperature anomalies on standard deviation (industry-quarter fixed effects)

	Standard deviation					
	(1)	(2)	(3)	(4)	(5)	(6)
Mean Headquarters	0.0011			0.0005		
	(1.4515)			(1.1267)		
Max Headquarters		0.0003			-1.498×10^{-7}	
		(0.9323)			(−0.0006)	
Min Headquarters			−0.0008**			−0.0003
			(−2.0117)			(−1.3135)
Mean Exchange × Different				0.0030*		
				(1.9322)		
Max Exchange × Different					0.0016*	
					(1.8600)	
Min Exchange × Different						−0.0022**
						(−2.0772)
Precipitation Headquarters	0.0002	0.0002	2.279×10^{-6}	0.0001	6.441×10^{-5}	-2.459×10^{-6}
	(1.4018)	(1.2405)	(0.0156)	(0.8872)	(0.5368)	(−0.0183)
Precipitation Exchange × Different				−0.0003	−0.0003	−0.0006
				(−0.4162)	(−0.3567)	(−0.7471)
VIX	-2.053×10^{-5}	-2.731×10^{-5}	-2.813×10^{-5}	-9.374×10^{-6}	-2.506×10^{-5}	-3.23×10^{-5}
	(−0.5623)	(−0.7899)	(−0.8246)	(−0.2364)	(−0.7082)	(−0.8956)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry-quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.0008	0.0003	0.0009	0.0035	0.0019	0.0036
No. observations	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007	1,963,007
No. firms	18,043	18,043	18,043	18,043	18,043	18,043

This table reports the effect of different temperature anomaly measures on the standard deviation of the returns of US firms with industry-quarter fixed effects. The dependent variable is the monthly standard deviation. The temperature variables are the absolute deviations of the monthly mean, maximum and minimum temperatures from the 30-year moving average of the same calendar month. Columns 4 to 6 include interaction terms between the exchange-city variables and Different. Different is a dummy for firms whose exchange is in a different city from their headquarters. t-statistics based on standard errors clustered by firm and by month are reported in parentheses below the coefficients. Significance stars are shown for the temperature anomaly variables: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.